# Lab 2 Submission Packet

Course Lab: Lab 2: Lerobot Ex ROS
Date: 2026-04-15

## 1. Rosetta Contract

```
robot_type: soa

fps: 30
max_duration_s: 15.0

observations:
  - key: observation.images.wrist
    topic: /follower/image_raw
    type: sensor_msgs/msg/Image
    image:
      resize: [480, 640]
    align: {strategy: hold, stamp: header}
    qos: {reliability: best_effort, history: keep_last, depth: 10}

  - key: observation.images.overhead
    topic: /static_camera/overhead_cam/color/image_raw
    type: sensor_msgs/msg/Image
    image:
      resize: [720, 1280]
    align: {strategy: hold, stamp: header}
    qos: {reliability: best_effort, history: keep_last, depth: 10}

  - key: observation.state
    topic: /follower/joint_states
    type: sensor_msgs/msg/JointState
    selector:
      names:
        - position.shoulder_pan
        - position.shoulder_lift
        - position.elbow_flex
        - position.wrist_flex
        - position.wrist_roll
        - position.gripper
    align: {strategy: hold, stamp: header}
    qos: {reliability: best_effort, history: keep_last, depth: 50}
    unit_conversion: rad2deg

actions:
  - key: action
    publish:
      topic: /follower/arm_fwd_controller/commands
      type: std_msgs/msg/Float64MultiArray
      qos: {reliability: reliable, history: keep_last, depth: 10}
    selector:
      names:
        - position.shoulder_pan
        - position.shoulder_lift
        - position.elbow_flex
        - position.wrist_flex
        - position.wrist_roll
    unit_conversion: rad2deg
    safety_behavior: hold

  - key: action
    publish:
      topic: /follower/gripper_fwd_controller/commands
      type: std_msgs/msg/Float64MultiArray
      qos: {reliability: reliable, history: keep_last, depth: 10}
    selector:
      names:
        - position.gripper
    unit_conversion: rad2deg
    safety_behavior: hold

recording:
  storage: mcap
```